Hyperspectral Image Classification Using Group-Aware Hierarchical Transformer

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***Abstract*— Hyperspectral image (HSI) classification is a critical task with numerous applications in the field of remote sensing. Although convolutional neural networks have achieved remarkable success in computer vision, they are still limited in the ability to model long-term dependencies due to small receptive fields. Recently, vision transformers have been used in HSI classification, where multi-head self-attention (MHSA), as the key feature extractor of transformers, learns global dependencies in long-range positions and bands of HSI pixels. Existing vision transformers for classifying HSIs with a large number of bands, however, have some limitations in that features extracted by MHSA may exhibit over-dispersion. In this article, we propose a Group-Aware Hierarchical Transformer (GAHT) for HSI classification, which confines MHSA to the local spatial–spectral context by introducing a new grouped pixel embedding (GPE) module. The GPE emphasizes local relationships within HSI spectral channels, resulting in a global–local fashion from a spatial–spectral context for HSI classification. In addition, we construct our transformer in a hierarchical manner, which can significantly improve classification accuracy with only a few parameters. Extensive experiments on four benchmark HSI datasets demonstrate that the proposed method outperforms state-of-the-art HSI classification algorithms. The source code is available at https://github.com/MeiShaohui/Group-Aware-Hierarchical-Transformer.**

***Index Terms*— Grouped pixel embedding (GPE), hierarchical transformer, hyperspectral image (HSI) classification, multi-head self-attention (MHSA).**

I. Introduction

With the rapid development of remote sensing and unmanned aerial vehicle (UAV) technology, the amount of available hyperspectral image (HSI) datasets has increased a lot. HSIs present abundant spectral and spatial information, where each pixel in an HSI reflects physical properties of materials within the pixel spatial coverage and can be used as "finger print" for material identification [1]. Therefore, HSI classification aims to recognize the category of each pixel, has numerous applications in remote sensing, such as precision agriculture [2], environmental monitoring [3], and mineral analysis [4].

Manuscript received 17 August 2022; revised 11 September 2022; accepted 15 September 2022. Date of publication 19 September 2022; date of current version 3 October 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62171381 and in part by the Fundamental Research Funds for the Central Universities. *(Corresponding author: Shaohui Mei.)*

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Digital Object Identifier 10.1109/TGRS.2022.3207933

Deep learning techniques have received a lot of attention in recent years [5] and are widely used in HSI classification areas [6], [7], [8], [9], which can be roughly classified into two main streams: spectral feature-based methods and spatial–spectral feature-based methods. Spectral feature-based methods simply feed each spectral pixel vector into classifiers. Without considering the spatial context of pixels, the performance of these methods is limited. While spatial–spectral feature-based methods integrate both spectral and spatial information of HSIs comprehensively. Specifically, spatial–spectral feature-based features are extracted by replacing the input with a patch, which is made of the target pixel and its neighbor pixels. Among spatial–spectral feature-based methods, convolutional neural networks (CNNs) have been widely used in HSI classification and demonstrated their effectiveness [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28]. Although CNN-based models have achieved promising classification results of HSIs, there still exist some problems. First, the receptive field of the CNN is restricted by small-size convolutional kernels (e.g., 3×3 or 5×5). Hence, these CNN-based algorithms are inferior in modeling long-range dependencies of global information in HSIs. Besides, complicated convolutions lead to difficulty in highlighting the importance of different spectral features.

Most recently, vision transformers have been widely used in various fields of computer vision, e.g., image classification, objection detection, and instance segmentation [32], [33], [34]. Vision transformers learn features mainly based on multi-head self-attention (MHSA) instead of convolutions. The MHSA mechanism can extract global relationship from nonoverlapping parts of the whole image. Hence, vision transformers have a superior capability in catching long-distance dependencies in the image. Moreover, the MHSA emphasizes prominent features while suppressing relatively insignificant features. Vision transformers based on the MHSA in conjunction with other components have approached or exceeded CNNs-based state-of-the-art (SOTA) performance [32], [33], [34]. As a break bottleneck, the vision transformer has also been applied for HSI classification and provides a new inspiration for HSI classification [43], [44], [52]. However, vision transformers are originally designed for natural images that are quite different from HSIs. Few modifications are made to fill this gap in existing transformers for classifying HSIs, resulting in the hindrance of making the HSI classification performance in a different league. The problems caused by the difference

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between this pair of heterologous images can be summarized as follows.

1) HSIs usually have hundreds of bands, while optical images only have three channels. When MHSA is applied to HSIs, the extracted feature may be over-dispersed in the global range due to abundant spectral information of HSIs. This ignores the local spectral information of HSIs, which also plays an important role in recognizing the category of ground objects.
2) Existing vision transformers for HSI classification are built in a nonhierarchical structure. Some advanced vision transformers have indicated the effectiveness of hierarchical design for processing natural images [33], [34]. For HSIs with a large number of spectral channels, models with nonhierarchical architectures render the problem of redundancy of features.

To overcome the above limitations of transformer-based methods for HSI classification, A novel end-to-end group-aware hierarchical transformer (GAHT) is proposed for classifying HSIs. Inspired by the technique of spectral partition [42] that emphasizes local representations in the spectral domain, we propose a novel and effective embedding module called grouped pixel embedding (GPE). The GPE models local contents as the complement of over-separated features extracted by the MHSA in the transformer. Furthermore, a hierarchical style of transformer network is constructed in the article. Compared with the original transformer, the hierarchical model can gradually reduce the channel size of the feature maps to extract multiscale features by a progressive shrinking manner, in which the computational cost is also significantly reduced. The contributions are summarized as follows.

1) We propose a simple yet effective embedding module named GPE to extract discriminative spatial–spectral features from nonoverlapping subchannels. By combining with MHSA, the attention can be confined to a local–global range, by which the drawback of over-dispersion of MHSA can be well handled.
2) A hierarchical structure is designed to explore multiscale Spatial–spectral features using the proposed transformer-based local–global attention mechanism, by which the parameters can be lessened while keeping competitive classification results.
3) Compared with several deep learning methods including some SOTA CNNs, experimental results demonstrate that the proposed network achieves superior classification accuracy on four HSI datasets.

The remainder of this article is organized as follows. Section II describes related work. In Section III, our proposed GAHT is introduced in detail. Section IV conducts extensive experiments on four HSI dataset to demonstrate the effectiveness of the proposed method. Finally, some conclusions are drawn in Section V.

II. Related Work

A. HSI Classification Using Spectral Feature-Based Methods

When deep learning was introduced for HSI classification, spectral feature-based methods are widely designed for the classification task, which extract vectorized features along the 1D spectral dimension. Hu *et al.* [10] employed deep CNNs to classify HSIs directly in the spectral domain. Chen *et al.* [11] presented a regularized deep feature extraction method for HSI classification using a CNN. Mou *et al.* [12] proposed a novel recurrent neural network (RNN) model that can effectively analyze hyperspectral pixels as sequential data to perform HSI classification. Wu and Prasad [13] used semisupervised deep learning by pseudo labels with a deep RNN. The performance of these spectral feature-based models is limited due to the absent of spatial context of pixels. Many simple attempts have been implemented into these models to further improve their performance. For example, composite kernels are applied to spatial neighborhood of pixels prior to 1-D CNN to improve its performance [14], [15]. However, these attempts treat spectral and spatial information separately, the increasement of accuracies is not remarkable.

B. Spatial–Spectral Feature-Based Methods for HSI Classification

Spectral feature-based methods ignore the spatial neighboring information of HSIs. Moreover, the integrity of the spatial–spectral features of HSIs is not fully considered. To overcome these problems, spatial–spectral feature-based methods, especially CNNs are widely used for HSI Classification. First, 2-D CNNs with 2-D convolutions and pooling kernels are introduced, which are capable of modeling the spatial information of HSIs. Makantasis *et al.* [16] proposed a 2-D CNN to construct high-level features in an automated way for classifying HSIs. Chen *et al.* [18] proposed a new strategy combines Gabor filters with convolutional filters to mitigate the problem of overfitting in HSI classification. Song *et al.* [19] proposed a novel model based on fusing mechanism by combining multiple-layer features to explore the complementary and related information from three different perspectives. Zhao *et al.* [20] proposed a dual-tunnel CNN to capture spectral and spatial features, which enforced spatial consistency in the deeper layers of the network. Mei *et al.* [21] proposed a new method to reduce memory requirements and accelerate the forward step of 2-D CNNs.

Considering the 3-D characteristics of HSIs, many 3-D CNNs based on 3-D convolutions are used to render the classification of HSIs. While the 3-D CNNs handle the spectral and spatial signatures simultaneously. Chen *et al.* [22] He *et al.* [23] proposed an end-to-end multiscale 3-D deep CNN for HSI classification, where 3-D method tries to capture both 2-D multiscale spatial feature and 1-D spectral feature of HSIs. Zhong *et al.* [24] constructed a spectral–spatial residual network to capture both spectral features and spatial features in sequential residual convolutional blocks. Hamida *et al.* [25] introduced a new 3-D approach that enables a joint spectral and spatial information process, meanwhile a set of 3-D schemes was proposed and evaluated in their work. Mei *et al.* [26] proposed an unsupervised spatial–spectral feature learning strategy, which allowed the 3-D convolutional autoencoder network to learn valid features without the need for pixel marker information.

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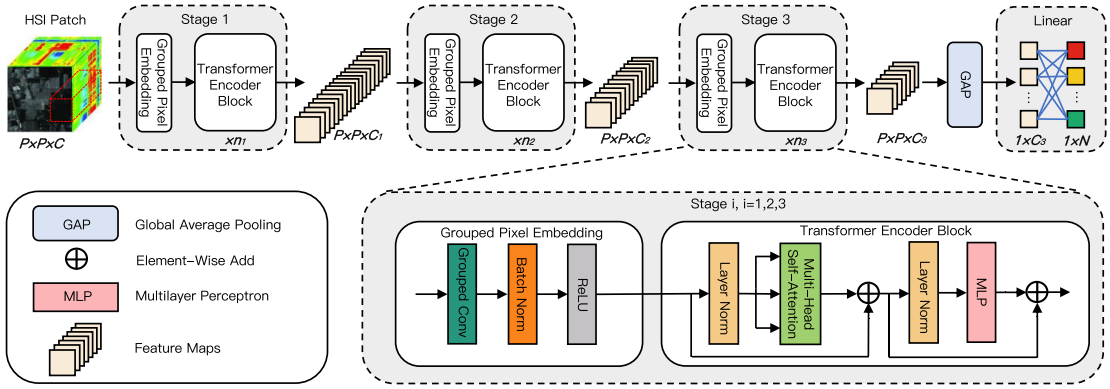


Fig. 1. Schematic of the proposed GAHT model for HSI classification. The core part of the network consists of three similar stages. Each stage owns one GPE module followed by several transformer encoder blocks. Transformer encoder blocks with the MHSA mechanism are used for modeling global dependencies in long-range positions and bands of a HSI patch. And GPE pays more attention to local relationships within Spatial–spectral features of HSIs. Moreover, the GPE in each stage is also used to reduce the dimension along the spectral axis, forming a hierarchical style of the transformer.

More recently, some work embeds the attention module into original networks to extract the correlation within spatial and spectral features in HSIs. These attention modules are stand-alone that can be flexibly plugged into the original HSIs classification algorithms. Sun *et al.* [36] proposed an attention mechanism to build an end-to-end spectral–spatial attention network to capture the correlation between the hyperspectral pixels within an HSI cube. Lu *et al.* [37] employed a novel 3-D channel and spatial attention-based multiscale spatial–spectral residual network. Zheng *et al.* [38] designed an attention mechanism to model the interdependencies of the feature maps with the spatial context. Zhu *et al.* [39] proposed an end-to-end residual spectral–spatial attention network (RSSAN) to select useful spectral bands and strength or weaken pixels in the neighborhood for HSI classification. Mei *et al.* [40] proposed an attention mechanism for selecting significant spatial context and spectral information of HSIs. Liu *et al.* [41] introduced a central attention network based on scaled dot-product central attention for HSI classification. Zhang *et al.* [42] proposed a novel spatial attention module named homogeneous pixel detection module and spectral partitioning residual network. Although these methods based on the attention have achieved promising classification results of HSIs, this kind of attention is still essentially convolution which suffers from the inherent drawbacks of it due to the fixed local structure of kernels. This results in the inferior representation of long-range interactions between spatial–spectral attributes. Moreover, the attention only highlights features of interest within local areas, which may ignore important relation among larger global spatial–spectral areas.

C. Vision Transformers for HSI Classification

Totally different from CNNs, vision transformers treat the image in a token style. These transformers usually cut the image into nonoverlapping patches. Each patch is treated as a token representing features of the corresponding position of the original image. By learning the relationship of these patches, vision transformers can describe better long-distance dependencies of features than CNNs. So vision transformers have been preliminarily introduced for HSI classification very recently. Hong *et al.* [43] proposed a backbone network based on the transformer, in which the attention mechanism is used to improve the ability to capture details of subtle spectral differences. Xue *et al.* [44] proposed a novel local transformer with the partial partition restore module. Sun *et al.* [52] constructed a spectral–spatial feature tokenization transformer (SSFTT) to capture spectral–spatial features and high-level semantic features. Compared to CNNs, we find that there is relatively little work on using vision transformers for HSI classification. So, vision transformers have great potential in the field of HSI classification.

III. Proposed Methodology

A. Overall Architecture

An overview of the proposed GAHT is shown in Fig. 1. The proposed GAHT follows a patch-based learning framework to explore spatial–spectral information of HSIs. As the key component of transformers, the MHSA mechanism is employed to calculate the internal correlation of global effective spatial–spectral features. Besides, the GPE module is proposed to divide the channels of feature maps into nonoverlapping subchannels and extract spatial–spectral features from these subchannels. Moreover, the three-stage hierarchical framework is designed to decrease the channel numbers of feature maps hierarchically to modeling low-level, middle-level, and high-level features for classification.

The proposed framework serves as an end-to-end learning manner. Given an input HSI patch cube $X \in \mathbb{R}^{P\times P\times C}$, where P is the spatial size of input HSI patch and C is the number of HSI bands. We feed the HSI patch to three sequential combinations of grouped pixel embedding and transformer encoder block called stages 1, 2, and 3 for learning spatial–spectral features in low, middle, and high level, respectively. These feature maps corresponding to the three stages are with the size of $P \times P \times C_1$, $P \times P \times C_2$, and $P \times P \times C_3$. The number of transformer blocks in each stage is denoted as n_1, n_2 and n_3. After passing through all stages, the feature map with size of $P \times P \times C_3$ will be applied by a global

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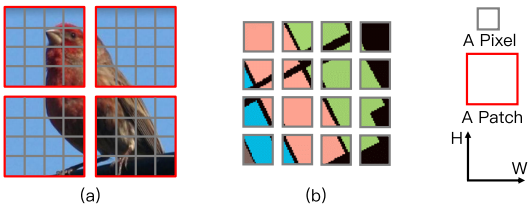


Fig. 2. Comparison of (a) patch embedding for optical images and (b) pixel embedding for HSIs.

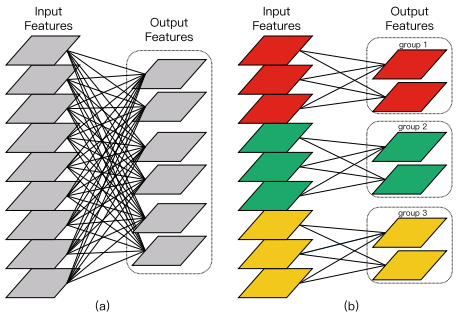


Fig. 3. Comparison of (a) normal 2-D convolutions and (b) grouped 2-D convolutions.

average pooling (GAP) to compress spatial information and generate a vector with a size of C_3 to determine the ground land label $y = 1, 2, \ldots, N$ of the center pixel in X. Finally, a fully connected layer is used to classify X, where N is the number of classes for a specific HSI dataset.

B. Grouped Pixel Embedding

The patch embedding module in vision transformers splits the image into a series of nonoverlapped patches and then projected them to feature maps. It is usually implemented by a 2-D convolution which has the same kernel size and stride. And the spatial size of generated image patches is identical to the kernel size of convolutions. For example, 64 convolution kernels which have the kernel size and stride of 32 divide an image with shape of $128 \times 128 \times 3$ into $((128/32) \times (128/32))$ patches with shapes of $32 \times 32 \times 64$. However, the original patch embedding is difficult to be directly applied for HSI classification. Specifically, the input of HSI classification algorithms is already a small patch of the HSI. Moreover, the spatial size of the input is usually odd (e.g., 7×7 or 9×9) which cannot be divided into nonoverlapped parts.

To solve these issues, we proposed pixel embedding module to replace standard patch embedding in vision transformers. As it literally means, the pixel embedding cut the HSI patch into isolated pixels. We simply use 1×1 convolutions split to the HSI patch into these pixel sequences. The 1×1 convolution can divide the HSI patch in a nonoverlapping way whether the size of the patch is odd or even. Moreover, the relationship between different pixels in a patch can be well learned by this split method. The principle of patch embedding and our pixel embedding is illustrated in Fig. 2.

Recently, grouped convolutions have been introduced in the field of HSI classification [42]. Fig. 3 shows the difference between normal convolutions and grouped convolutions. Specifically, for an input feature map $X \in \mathbb{R}^{H \times W \times C}$, where H and W are the height and width of the feature map, and C is the number of channels. X is first divided into n nonoverlapping feature maps along the channel dimension

$$X = \{X_1, X_2, \ldots, X_i, \ldots, X_n\} \tag{1}$$

where n is the number of groups, and $X_i \in \mathbb{R}^{H \times W \times (C/n)}$. Then different convolutions are applied on every X_i

$$X_i' = W_i X_i + b_i, \quad i = 1, 2, \ldots, n \tag{2}$$

where X_i' is the output of the ith convolution, W_i is ith weight matrix defined by the filter bank with kernel size $k \times k$, and b_i is the bias of the ith convolution. Finally, the features are concatenated to generate the output of grouped convolution denoted as X'

$$X' = \text{Concat}(X_1, X_2, \ldots, X_i, \ldots, X_n). \tag{3}$$

Compared with common convolutions, grouped convolutions have fewer parameters and is not easy to overfit. Meanwhile, grouped convolutions can be used to extract discriminative information shared by continuous subchannels of feature maps. Hence, grouped convolutions are better at capturing local information of feautures. We further construct our pixel embedding in this grouped style to generate GPE. Existing methods show that imposing batch normalization (BN) [48] and rectified linear unit (ReLU) activation on every convolutional layer can regularize the learning process and improve the classification performance [24]. Hence, we also place a BN layer and ReLU function after each grouped convolution to construct the GPE module. which are denoted as

$$X' = \text{BN}(X) = \gamma \frac{X - \mu}{\sigma} + \beta \tag{4}$$

$$\text{ReLU}(X) = \max(0, X) \tag{5}$$

where X and X' are batches of the input and output fed into the BN layer, while μ and σ are the expectation and variance of one batch. γ and β are parameters to be learned during the training process. To sum up, our proposed GPE can be written as

$$\text{GPE}(X) = \text{ReLU}\left(\text{BN} f_{1\times1}^{g}(X)\right) \tag{6}$$

where $f_{1\times1}^{g}$ stands for the 1×1 grouped convolution.

Finally, to construct a hierarchical framework, the out_channels is half of the in_channles in grouped convolutions, which means $C_3 = (1/2)C_2 = (1/4)C_1$ in Fig. 1. With the designed hierarchical structure, resource consumption can be reduced and useful features can be mined.

C. Transformer Encoder Block

In the proposed GAHT, the transformer encoder block consists of alternating layers of an MHSA module and a multilayer perceptron (MLP) layer.

MHSA [35] is a mechanism aiming at building a globe long-range dependencies and extracting important parts of features. MHSA linearly projects the input into multiple feature subspaces and processes them by several independent

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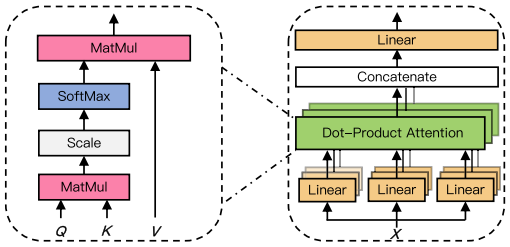


Fig. 4. Structure of MHSA mechanism.

TABLE I
NUMBER OF TRAINING, VALIDATION AND TESTING SAMPLES FOR THE SA DATASET

No.	Name	Train.	Val.	Test.
1	Brocoli_green_weeds_1	40	40	1929
2	Brocoli_green_weeds_2	74	75	3577
3	Fallow	40	39	1897
4	Fallow_rough_plow	28	28	1338
5	Fallow_smooth	53	54	2571
6	Stubble	79	79	3801
7	Celery	71	72	3436
8	Grapes_untrained	225	226	10820
9	Soil_vinyard_develop	124	124	5955
10	Corn_senesced_green_weeds	65	66	3147
11	Lettuce_romaine_4wk	22	21	1025
12	Lettuce_romaine_5wk	39	38	1850
13	Lettuce_romaine_6wk	19	18	879
14	Lettuce_romaine_7wk	22	21	1027
15	Vinyard_untrained	145	146	6977
16	Vinyard_vertical_trellis	36	36	1735
	Total	1082	1083	51964

TABLE II
NUMBER OF TRAINING, VALIDATION AND TESTING SAMPLES FOR THE PU DATASET

No.	Name	Train.	Val.	Test.
1	Asphalt	132	133	6366
2	Meadows	373	373	17903
3	Gravel	42	42	2015
4	Trees	62	61	2941
5	Painted metal sheets	27	27	1291
6	Bare Soil	100	101	4828
7	Bitumen	27	26	1277
8	Self-Blocking Bricks	73	74	3535
9	Shadows	19	19	909
	Total	855	856	41065

TABLE III
NUMBER OF TRAINING, VALIDATION AND TESTING SAMPLES FOR THE WHU-LK DATASET

No.	Name	Train.	Val.	Test.
1	Corn	172	173	34166
2	Cotton	42	42	8290
3	Sesame	15	15	3001
4	Broad-leaf soybean	316	316	62580
5	Narrow-leaf soybean	21	21	4109
6	Rice	59	60	11735
7	Water	335	335	66386
8	Roads and houses	36	35	7053
9	Mixed weed	26	26	5177
	Total	1022	1023	202497

attention heads. The resulting vectors are concatenated and mapped to the final output. Like convolutions, MHSA enriches the diversity of feature subspace without extra calculation time by parallel computation. The process of MHSA, as shown in Fig. 4, can be formulated as the following steps. First, we generate a 2-D input matrix $X \in \mathbb{R}^{s\times d}$ by squeezing the spatial dimension of input features. Then, the input 2-D sequence X is mapped to three different matrices including query $Q \in \mathbb{R}^{s\times d}$, key $K \in \mathbb{R}^{s\times d}$, and value $V \in \mathbb{R}^{s\times d}$ by linear transformations as follows:

$$Q = XW^Q, K = XW^K, V = XW^V \tag{7}$$

where s and d are the length and dimension of the input sequence, respectively, the s is equal to the square of patch size for an HSI patch while d is the number of channels. And W^Q, W^K, and W^V are d-by-d matrices. Then, Q, K, and V are split by n along the dimension axis as follows:

$$\begin{aligned} Q &= \{Q_1, Q_2, \ldots, Q_i, \ldots, Q_n\} \\ K &= \{K_1, K_2, \ldots, K_i, \ldots, K_n\} \\ V &= \{V_1, V_2, \ldots, V_i, \ldots, V_n\} \end{aligned} \tag{8}$$

where n is the number of heads, $Q_i, K_i, V_i \in \mathbb{R}^{s\times(d/n)}$. Next, the attention score for every i is computed by a scaled softmax function as

$$Z_i = \text{Attention}(Q_i, K_i, V_i) = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d}}\right)V_i. \tag{9}$$

The final output of MHSA is generated by a projection after concatenating all $Z_i \in \mathbb{R}^{s\times(d/n)}$, which can be formulated as

$$\text{Output} = \text{Concat}(Z_1, Z_2, \ldots, Z_i, \ldots, Z_n)W^O \tag{10}$$

where $W^O \in \mathbb{R}^{d\times d}$ denotes the output projection matrix.

The MLP is a 2-layer linear projection with a Gaussian error linear unit (GELU) nonlinearity [49] in between to extract feature representation.

The linear projection in MLP is simply implemented by a fully connected (FC) layer. And the GELU is a commonplace function widely used in vision transformer, which is defined as

$$\text{GELU}(X) = X\Phi(X) = \frac{X}{2}\left[1 + \text{erf}\left(\frac{X}{\sqrt{2}}\right)\right] \tag{11}$$

where $\Phi(X)$ is the standard Gaussian cumulative distribution function, and $\text{erf}(X) = \int_0^X e^{-t^2}dt$. The MLP module can be summarized as

$$\text{MLP}(X) = \text{FC}_2(\text{GELU}(\text{FC}_1(X))). \tag{12}$$

In transformer block, LayerNorm [50] layers are applied before an MHSA module and an MLP layer. Moreover,

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TABLE IV

NUMBER OF TRAINING, VALIDATION AND TESTING SAMPLES FOR THE HR-L DATASET

No.	Name	Train.	Val.	Test.
1	Dense Urban Fabric	8	9	271
2	Mineral Extraction Sites	2	2	63
3	Non Irrigated Arable Land	17	16	509
4	Fruit Trees	3	2	74
5	Olive Groves	42	42	1317
6	Broad-leaved Forest	6	7	210
7	Coniferous Forest	15	15	470
8	Mixed Forest	32	32	1008
9	Dense Sclerophyllous Vegetation	114	114	3565
10	Sparce Sclerophyllous Vegetation	84	84	2635
11	Sparcely Vegetated Areas	12	12	380
12	Rocks and Sand	14	15	458
13	Water	42	42	1309
14	Coastal Water	14	13	424
	Total	405	405	12693

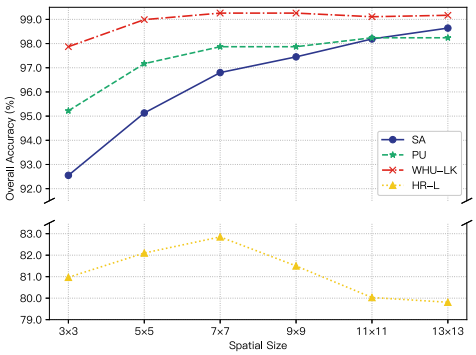


Fig. 5. OA (%) with different spatial sizes on the four datasets.

TABLE V

OVERALL ACCURACIES (%) OF PROPOSED METHOD WITH DIFFERENT GROUP NUMBERS IN GPE MODULE ON THE SA, PU, WHU-LK, AND HR-L DATASETS

Dataset	2	4	8	16	32
SA	96.26	96.19	96.23	**96.33**	96.18
PU	**97.61**	97.40	97.53	97.06	96.73
WHU-LK	**99.25**	99.17	99.09	99.16	99.08
HR-L	81.92	**82.38**	82.02	81.78	82.08

residual connections [51] are adopted to alleviate the problem of overfitting which is common for deep learning methods. The overall computation of consecutive transformer encoder blocks can be summarized as

$$\widetilde{z}_l = \text{MHSA}(\text{LayerNorm}(z_l)) + z_l \tag{13}$$

$$z_{l+1} = \text{MLP}\big(\text{LayerNorm}\big(\widetilde{z}_l\big)\big) + \widetilde{z}_l \tag{14}$$

where $\widetilde{z}_l$ and z_l denote the output features of the MHSA module and MLP for block l, respectively. The number of transformer blocks in each stage varies according to the results of experiment, which will be discussed in Section IV.

TABLE VI

OVERALL ACCURACIES (%) OF PROPOSED METHOD WITH DIFFERENT NUMBER OF TRANSFORMER BLOCKS IN EACH STAGE ON THE SA, PU, WHU-LK, AND HR-L DATASETS

Depth	SA	PU	WHU-LK	HR-L
1+1+1	96.33	97.61	99.25	82.38
2+1+1	**96.80**	97.65	99.18	82.41
1+2+1	96.79	**97.87**	99.21	**82.85**
1+1+2	96.55	97.61	99.19	82.13
1+2+2	99.14	97.64	99.14	82.69
2+1+2	96.51	97.70	99.21	81.79
2+2+1	96.61	97.63	**99.26**	82.49
2+2+2	96.17	97.64	99.21	82.04

IV. EXPERIMENTS

To evaluate the performance of our proposed method, we conduct extensive experiments to demonstrate the effectiveness of the proposed method on four benchmark datasets including two classic ones [i.e., Salinas (SA) and Pavia University (PU)] and two recently released ones [i.e., Wuhan UAV-borne hyperspectral image-LongKou (WHU-LK) and HyRANK-Loukia (HR-L)].

A. Datasets

Here, four datasets used in the experiments are introduced in detail. The false-color composite images of datasets are shown as subfigures within Figs. 6–9

1) Salinas: The SA dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1998. The original image has 224 bands with wavelengths ranging from 400 to 2500 nm. After deleting the water absorption bands, 204 bands are employed for evaluating. The height and width of the data are 512 and 217 pixels, respectively. The dataset has 54129 marked samples presenting 16 object classes.

2) Pavia University: The PU dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in 2001. It includes 115 spectral bands ranging from 380 to 860 nm. After the noise bands were discarded, 103 available bands are used for studying. The image has a height of 610 and a width of 340 pixels, respectively. There are a total of 42 776 labeled samples of 9 land covers in the dataset.

3) WHU-LongKou: The WHU-LK dataset was acquired in Longkou Town, Hubei province, China, with an 8-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJ-innovations (DJI) Matrice 600 Pro UAV platform [46], [47]. It consists of 270 spectral bands with the wavelength range of 400 to 1000 nm. The size of the imagery is 550 × 400 pixels. This dataset has 204542 marked samples presenting nine different object classes.

4) HyRANK-Loukia: The HR-L dataset is one of five satellite HSIs from the Earth Observing-1 (EO-1) Hyperion sensor in the framework of the International Society for Photogrammetry and Remote Sensing (ISPRS) scientific initiatives [45]. The HR-L image consists of 249 × 945 pixels and 176 spectral bands. In this dataset, a total number of 13503 pixels is manually labeled and 14 different land covers are included.

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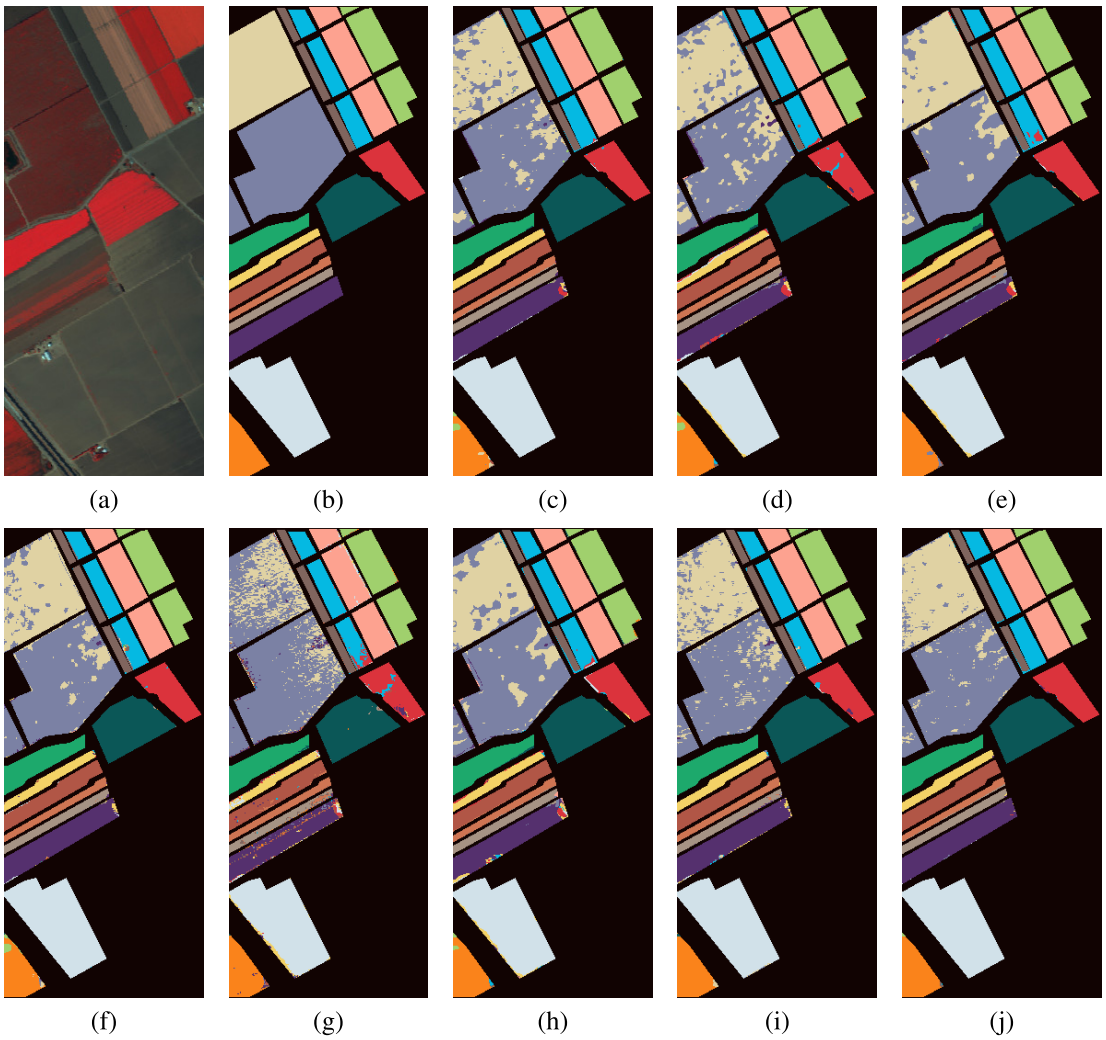


Fig. 6. Classification maps of the SA dataset with 1% training samples. (a) False-Color Image. (b) Ground Truth. (c) M3D-DCNN. (d) 3D-CNN. (e) DFFN. (f) RSSAN. (g) AB-LSTM. (h) SF. (i) SSFTT. (j) Proposed GAHT.

We used different proportions of label samples for each dataset. For the SA and PU datasets, we randomly select 1% labeled samples for training, 1% are used for validation, and 98% are used for testing. For the WHU-LK dataset, the proportions are 0.5%, 0.5%, and 99%. And for the HR-L dataset, 3%, 3%, and 94% of labeled samples are selected for training, validation, and testing, respectively. The details of the number of each class for training, validation, and testing sets of four datasets can be found in Tables I–IV.

B. Experiment Setups

To make a fair comparison, all the HSI classification methods are implemented on a deep learning framework of PyTorch 1.8.2. The minibatch size and the epoch are set to 64 and 300 to update the training parameters in all frameworks, respectively. To get the best performance of other models reported in their articles, the optimizer and learning rate scheduler are the same as their original settings. For our method, we employ the stochastic gradient decent (SGD) optimizer with a momentum of 0.9 and a weight decay of 0.0001 to update the training parameters. We do not use any learning rate scheduler, and the learning rate is a constant of 0.001 for the sake of simplicity. We used an NVIDIA GeForce RTX 3080 graphics processing unit (GPU) with 10-GB memory to accelerate the experiments. For settings of transformer architecture, the embedding dimension for three stages $\{C_1, C_2, C_3\}$ is set as $\{256, 128, 64\}$ in our implementation. And the number of heads in the MHSA for different stages is $\{8, 4, 2\}$. More details of our proposed method can be found in the source code mentioned in the abstract.

To achieve steady training progress for deep neural networks, we employ the data preprocessing method which introduced in [36]. Let $X \in R^{B\times H\times W}$ represent the original HSI data, where B denotes the band number of spectral domain, while H and W denote the spatial height and weight. First, all data values are normalized to the range [0, 1] by

$$X' = \frac{X - \min(X)}{\max(X) - \min(X)}. \tag{15}$$

Then, we normalize the HSI data in every spectral band by the average value of the corresponding channel

$$X''(b, :, :) = X'(b, :, :) - \text{mean}\big(X'(b, :, :)\big) \tag{16}$$

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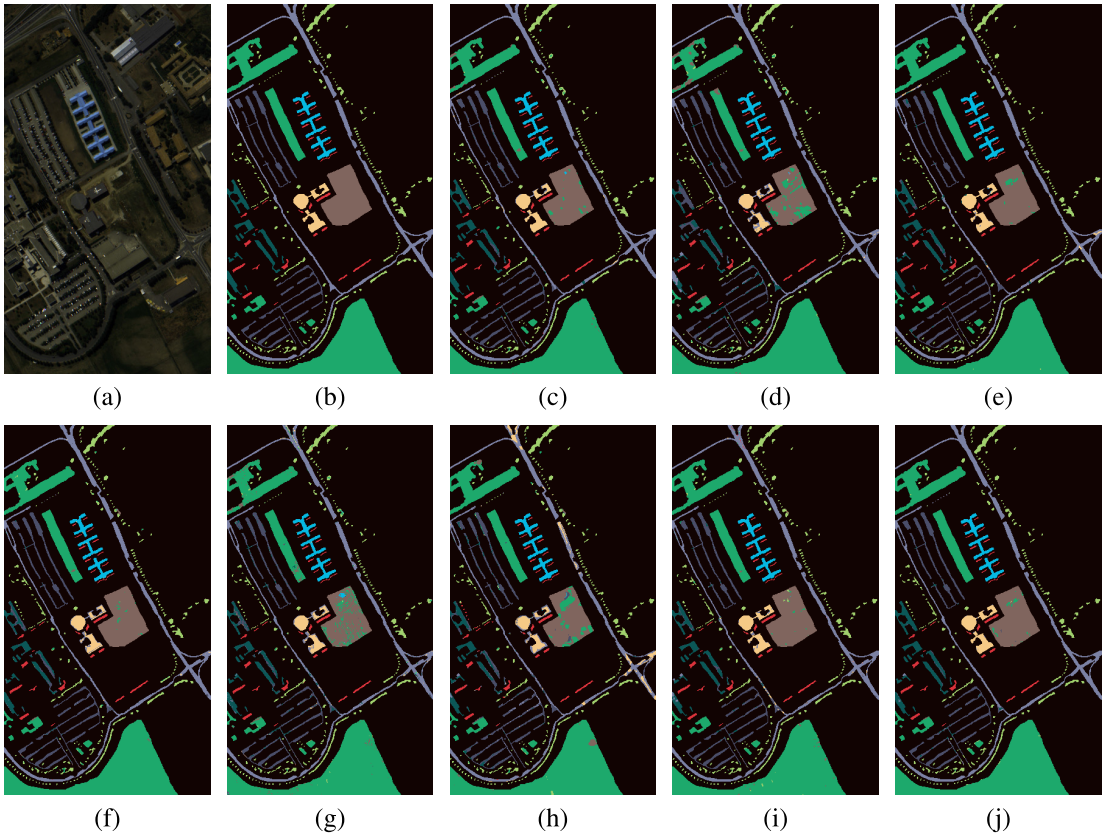


Fig. 7. Classification maps of the PU dataset with 1% training samples. (a) False-Color Image. (b) Ground Truth. (c) M3D-DCNN. (d) 3D-CNN. (e) DFFN. (f) RSSAN. (g) AB-LSTM. (h) SF. (i) SSFTT. (j) Proposed GAHT.

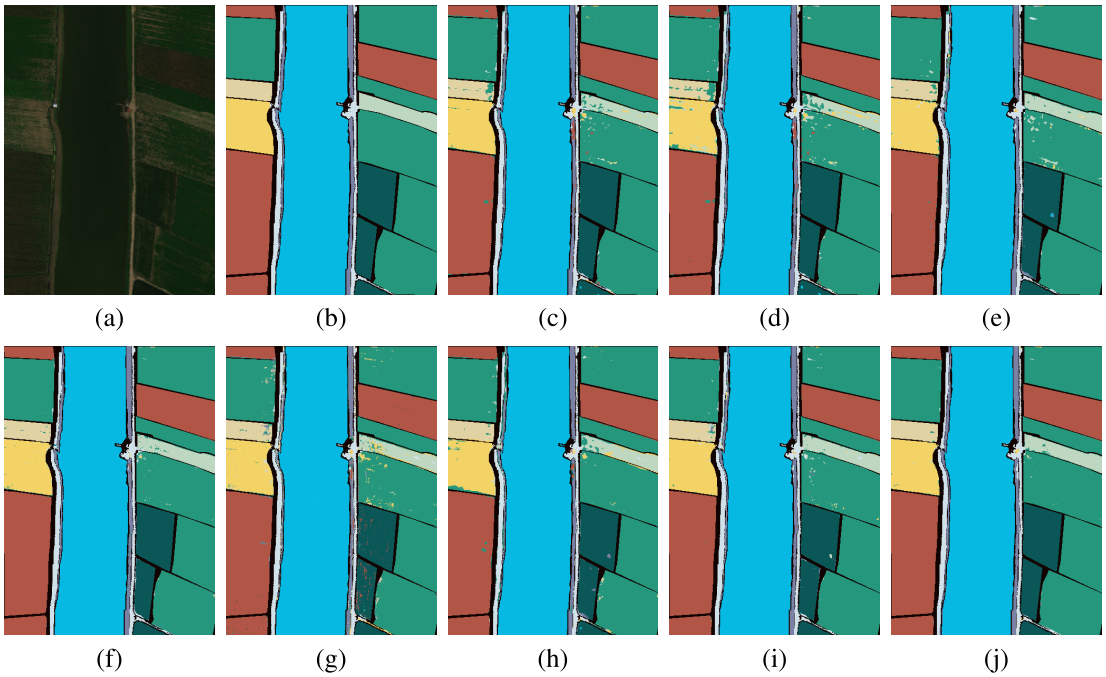


Fig. 8. Classification maps of the WHU-LongKu dataset with 0.5% training samples. (a) False-Color Image. (b) Ground Truth. (c) M3D-DCNN. (d) 3D-CNN. (e) DFFN. (f) RSSAN. (g) AB-LSTM. (h) SF. (i) SSFTT. (j) Proposed GAHT.

where $\min(X)$ and $\max(X)$ represent the minimum and maximum of X, respectively. And $\text{mean}(X'(b,:,:))$ denotes the mean value of bth channel in a processed HSI X'.

Also, the data augmentation is utilized to alleviate the problem of insufficient train samples for all methods. To be specific, for every batch of data fed in one iteration, we

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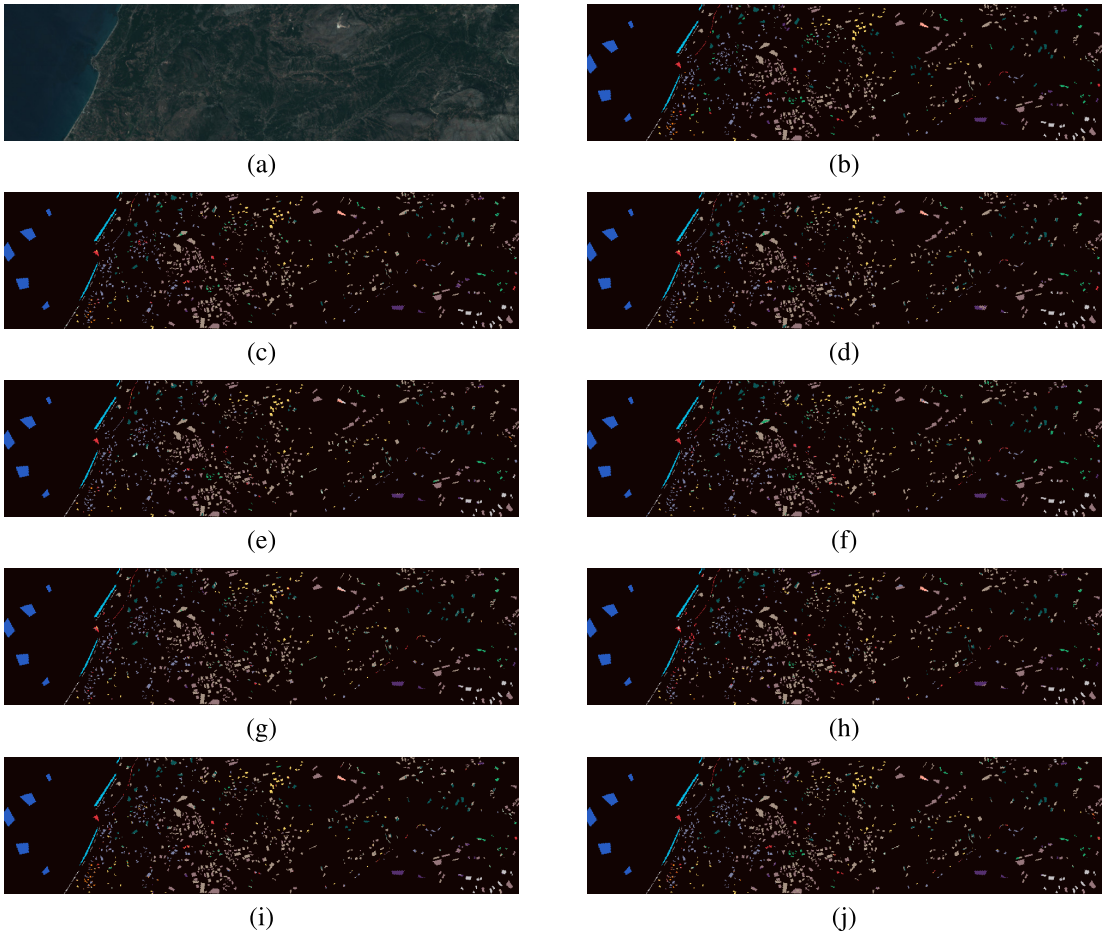


Fig. 9. Classification maps of the HR-L dataset with 3% training samples. (a) False-Color Image. (b) Ground Truth. (c) M3D-DCNN. (d) 3D-CNN. (e) DFFN. (f) RSSAN. (g) AB-LSTM. (h) SF. (i) SSFTT. (j) Proposed GAHT.

randomly select one of the following five data augmentation methods with the same probability: vertical flip, horizontal flip, 90° rotation, 180° rotation, and 270° rotation.

In order to judge the HSI classification performance quantitatively, we employ the overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and each-class accuracies as evaluation indicators. OA represents the number of correctly predicted samples of the total number of samples, AA is the average of the classification accuracy of each class, and κ is a statistic measurement evaluating the consistence of the ground truth map and classification map. To reduce the impact of experimental randomness, we randomly split the labeled samples five times and report their mean values and standard deviations. Moreover, the classification color maps are visualized to qualitatively compare the results obtained by different methods.

C. Parameter Analysis

1) Size of Spatial Input of HSIs: The spatial input size decides how much local spatial information is used for classifying a pixel in an HSI. So, the spatial size of HSI patches usually has a great influence on the performance of classification. In this section, we consider a set of spatial sizes {3, 5, 7, 9, 11, 13} to explore the effects of spatial sizes. As shown in Fig. 5, the classification performance of the proposed GAHT generally increases when the spatial size becomes larger. However, the accuracy of classification does not always be better as the spatial size increases. When the space sizes are 11 and 9, the OA of GAHT begins to drop on the Pavia University and WHU-LK datasets, respectively. The main reason is that the status of regions in each HSIs is different. For example, the SA dataset has larger smooth regions, the OA value generally gains as the spatial size becomes larger. We can find that the image of HR-L has small and separate regions of land covers. It explains when the patch size is greater than 7, the OA of the dataset drops dramatically. For a fair comparison, the patch size of HSI is set to 7×7 for all methods and datasets.

2) Number of Groups in Proposed GPE Module: The group number in grouped convolutions in the proposed GPE module is an important parameter, it decides the number of divided subchannels in features to be convoluted. So in this section, we analyze the performance of GAHT with a different number of groups to find the best settings for four different datasets. We simply fix the number of transformer block in every stage to be 1. Table V records the OA values of the experiment. From Table V, we can conclude that the best number of groups varies in different datasets. Specifically, for the PU and WHU-LK datasets, the best group number is 2. And for the SA and HR-L datasets, the best OA appears when the group numbers are equal to 16 and 4, respectively. It is also found that a higher number of groups does not always result in better result. In the

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TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE SA DATASET WITH 1% TRAINING SAMPLES

Class	M3D-DCNN	3D-CNN	DFFN	RSSAN	AB-LSTM	SF	SSFTT	GAHT
1	98.91 ± 1.19	97.07 ± 0.81	97.14 ± 1.85	99.42 ± 0.71	94.36 ± 7.76	95.71 ± 1.24	**100.00 ± 0.00**	99.84 ± 0.28
2	99.98 ± 0.03	99.86 ± 0.07	99.80 ± 0.14	99.28 ± 0.48	98.39 ± 0.92	99.48 ± 0.26	99.66 ± 0.52	**100.0 ± 0.00**
3	95.76 ± 2.50	92.14 ± 2.20	95.34 ± 2.32	98.31 ± 1.87	88.49 ± 4.19	95.19 ± 1.28	96.73 ± 1.71	**98.99 ± 0.91**
4	97.47 ± 1.57	98.33 ± 0.62	97.09 ± 1.21	98.84 ± 1.05	92.96 ± 11.13	94.55 ± 2.30	**99.22 ± 0.61**	98.90 ± 1.34
5	98.19 ± 1.02	96.23 ± 0.97	95.58 ± 3.07	96.43 ± 2.48	96.14 ± 3.08	92.20 ± 4.00	96.59 ± 2.68	**98.88 ± 1.26**
6	99.75 ± 0.23	99.58 ± 0.40	99.79 ± 0.37	98.55 ± 1.72	98.03 ± 1.73	99.42 ± 0.58	99.69 ± 0.37	**100.0 ± 0.00**
7	99.83 ± 0.06	99.38 ± 0.21	99.53 ± 0.22	99.09 ± 0.85	99.03 ± 0.42	98.25 ± 0.70	99.70 ± 0.21	**99.91 ± 0.11**
8	86.55 ± 2.55	83.96 ± 1.84	90.19 ± 2.92	89.02 ± 1.85	80.59 ± 10.63	84.97 ± 2.03	87.59 ± 1.46	**93.18 ± 1.31**
9	98.87 ± 1.64	97.89 ± 1.25	97.67 ± 1.75	99.51 ± 0.34	97.98 ± 1.53	96.76 ± 1.44	98.91 ± 1.22	**99.97 ± 0.04**
10	92.96 ± 1.55	89.27 ± 1.81	91.96 ± 0.83	94.47 ± 2.00	83.41 ± 1.25	89.64 ± 2.80	94.51 ± 1.71	**96.88 ± 1.50**
11	97.42 ± 1.98	83.08 ± 9.07	96.37 ± 3.19	96.62 ± 1.99	62.71 ± 32.59	90.05 ± 1.77	96.70 ± 2.25	**99.06 ± 0.95**
12	99.34 ± 0.59	99.06 ± 0.65	99.74 ± 0.35	99.60 ± 0.22	97.70 ± 1.86	98.47 ± 1.38	**99.88 ± 0.07**	99.84 ± 0.20
13	97.35 ± 3.16	97.62 ± 2.79	**98.46 ± 2.69**	96.28 ± 5.42	97.39 ± 1.42	92.61 ± 7.78	97.84 ± 2.66	96.70 ± 4.24
14	96.34 ± 2.01	96.36 ± 2.57	97.03 ± 1.71	96.51 ± 4.38	70.49 ± 33.40	96.57 ± 2.67	91.80 ± 4.85	**98.07 ± 1.40**
15	83.67 ± 3.78	74.55 ± 5.08	85.34 ± 5.37	86.65 ± 1.84	42.38 ± 23.59	79.07 ± 4.88	81.42 ± 3.77	**90.82 ± 2.01**
16	94.57 ± 3.13	91.76 ± 3.17	95.87 ± 2.88	96.13 ± 2.17	89.03 ± 7.16	92.72 ± 4.21	**97.09 ± 1.58**	96.87 ± 3.03
OA (%)	93.70 ± 0.35	90.89 ± 0.51	94.30 ± 0.43	94.68 ± 0.63	83.85 ± 1.66	91.32 ± 0.78	93.73 ± 0.39	**96.80 ± 0.35**
AA (%)	96.06 ± 0.29	93.51 ± 0.62	96.06 ± 0.29	96.54 ± 0.60	86.82 ± 3.08	93.48 ± 0.73	96.08 ± 0.47	**98.00 ± 0.24**
$\kappa\times100$	92.99 ± 0.39	89.86 ± 0.57	93.65 ± 0.48	94.08 ± 0.70	81.96 ± 1.90	90.34 ± 0.88	93.01 ± 0.44	**96.44 ± 0.39**

TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE PU DATASET WITH 1% TRAINING SAMPLES

Class	M3D-DCNN	3D-CNN	DFFN	RSSAN	AB-LSTM	SF	SSFTT	GAHT
1	95.69 ± 2.05	94.06 ± 2.04	97.08 ± 1.70	96.79 ± 1.87	90.75 ± 2.14	86.31 ± 2.55	96.16 ± 0.94	**98.11 ± 0.44**
2	98.97 ± 0.42	96.76 ± 1.58	99.27 ± 0.50	99.26 ± 0.69	97.86 ± 0.80	97.05 ± 1.12	99.29 ± 0.20	**99.59 ± 0.20**
3	86.00 ± 5.20	51.91 ± 8.88	85.72 ± 3.75	79.96 ± 4.95	66.95 ± 6.54	70.06 ± 7.53	83.99 ± 2.21	**88.15 ± 9.36**
4	96.10 ± 0.78	95.76 ± 1.30	95.85 ± 1.77	93.91 ± 0.97	91.91 ± 2.66	91.94 ± 1.58	**97.74 ± 0.79**	97.46 ± 0.98
5	99.94 ± 0.07	99.47 ± 0.67	99.44 ± 0.73	97.78 ± 1.60	98.21 ± 1.17	**100.00 ± 0.00**	99.95 ± 0.06	**100.00 ± 0.00**
6	90.20 ± 7.11	67.65 ± 9.78	95.45 ± 2.93	95.75 ± 2.72	80.15 ± 2.60	78.41 ± 4.06	97.63 ± 1.19	**97.96 ± 1.08**
7	88.23 ± 4.99	68.20 ± 5.38	84.07 ± 10.94	84.27 ± 6.73	80.84 ± 3.83	62.47 ± 2.92	88.84 ± 6.40	**92.00 ± 7.08**
8	92.60 ± 3.00	90.29 ± 1.74	91.95 ± 3.79	85.64 ± 9.71	86.85 ± 2.95	78.73 ± 8.10	93.50 ± 1.74	**95.33 ± 2.55**
9	99.01 ± 0.58	99.53 ± 0.38	99.46 ± 0.30	98.71 ± 0.99	98.99 ± 0.48	95.93 ± 0.96	99.81 ± 0.14	**99.83 ± 0.05**
OA (%)	95.74 ± 1.27	89.35 ± 1.88	96.48 ± 0.59	95.44 ± 0.69	91.29 ± 1.07	88.92 ± 1.07	96.96 ± 0.28	**97.87 ± 0.37**
AA (%)	94.08 ± 1.38	84.85 ± 2.18	94.25 ± 1.43	92.45 ± 0.78	88.06 ± 4.28	84.54 ± 1.37	95.21 ± 0.75	**96.49 ± 0.96**
$\kappa\times100$	94.33 ± 1.71	85.69 ± 2.58	95.32 ± 0.78	93.94 ± 0.92	88.37 ± 1.43	85.21 ± 1.46	95.97 ± 0.38	**97.18 ± 0.50**

TABLE IX
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE WHU-LK DATASET WITH 0.5% TRAINING SAMPLES

Class	M3D-DCNN	3D-CNN	DFFN	RSSAN	AB-LSTM	SF	SSFTT	GAHT
1	99.42 ± 0.33	99.30 ± 0.25	99.63 ± 0.15	99.55 ± 0.38	98.28 ± 2.01	99.58 ± 0.16	99.84 ± 0.09	**99.92 ± 0.03**
2	93.94 ± 2.57	90.62 ± 2.71	94.21 ± 1.68	96.77 ± 0.96	91.25 ± 3.13	91.85 ± 3.05	97.98 ± 0.94	**98.15 ± 0.84**
3	86.02 ± 6.40	77.11 ± 4.35	93.81 ± 2.69	96.67 ± 1.71	80.49 ± 7.02	94.31 ± 1.06	93.34 ± 1.70	**96.97 ± 1.88**
4	98.50 ± 0.58	98.64 ± 0.28	98.82 ± 0.32	99.22 ± 0.12	98.20 ± 0.76	98.12 ± 0.56	99.11 ± 0.15	**99.45 ± 0.22**
5	86.39 ± 5.63	72.88 ± 5.89	77.28 ± 17.31	91.48 ± 1.76	81.61 ± 2.96	83.88 ± 3.61	90.10 ± 2.93	**92.71 ± 3.39**
6	98.47 ± 1.10	99.20 ± 0.36	99.02 ± 0.71	98.61 ± 1.06	95.82 ± 1.33	97.08 ± 1.58	99.43 ± 0.53	**99.87 ± 0.11**
7	99.97 ± 0.04	99.97 ± 0.04	99.93 ± 0.07	99.98 ± 0.02	99.96 ± 0.03	99.96 ± 0.03	**99.99 ± 0.01**	99.94 ± 0.04
8	94.97 ± 2.15	93.45 ± 2.81	95.12 ± 1.55	96.55 ± 1.66	92.83 ± 1.84	89.52 ± 3.92	96.24 ± 1.50	**97.25 ± 1.16**
9	88.14 ± 4.36	82.53 ± 4.81	85.67 ± 5.65	91.87 ± 2.86	68.59 ± 8.66	75.43 ± 4.53	**94.30 ± 2.43**	93.37 ± 2.28
OA (%)	98.13 ± 0.29	97.46 ± 0.18	98.17 ± 0.47	98.91 ± 0.14	96.83 ± 0.39	97.43 ± 0.34	99.00 ± 0.10	**99.26 ± 0.19**
AA (%)	93.98 ± 0.89	90.41 ± 1.09	93.72 ± 2.22	96.74 ± 0.39	89.67 ± 1.14	92.19 ± 0.80	96.70 ± 0.53	**97.52 ± 0.81**
$\kappa\times100$	97.54 ± 0.38	96.65 ± 0.24	97.59 ± 0.62	98.57 ± 0.18	95.82 ± 0.51	96.61 ± 0.45	98.69 ± 0.13	**99.02 ± 0.25**

following experiments, we set the best group numbers reported in Table V for every HSI dataset.

3) Number of Transformer Blocks in Each Stage: The number of transformer blocks in each stage also is an important parameter. Since the three stages extract low-level, middle-level, and high-level features, respectively, we do not know in advance what kind of level of features are more suitable for classification in a certain dataset. Based on this, we determined the best combination through the experiment. Considering the capacity of the model, we limit each stage to own a maximum of two transform blocks, so there are eight different permutations in total. In this section, we explore the performance of

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TABLE X
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE HR-L DATASET WITH 3% TRAINING SAMPLES

Class	M3D-DCNN	3D-CNN	DFFN	RSSAN	AB-LSTM	SF	SSFTT	GAHT
1	54.39 ± 8.49	39.78 ± 10.78	37.71 ± 17.86	49.37 ± 15.29	35.13 ± 20.10	47.75 ± 9.05	61.33 ± 7.92	**65.54 ± 9.17**
2	90.79 ± 12.55	89.84 ± 10.36	79.68 ± 16.35	61.27 ± 26.00	70.48 ± 11.47	**98.41 ± 3.17**	65.08 ± 19.47	80.63 ± 19.22
3	82.28 ± 3.25	75.21 ± 3.83	80.16 ± 4.35	82.08 ± 2.57	80.04 ± 7.59	66.44 ± 6.55	82.08 ± 4.79	**87.66 ± 3.65**
4	30.54 ± 11.36	5.41 ± 4.60	4.86 ± 7.33	14.59 ± 9.10	0.00 ± 0.00	1.89 ± 2.36	**40.54 ± 7.50**	25.14 ± 15.01
5	85.92 ± 3.12	75.02 ± 3.29	88.93 ± 3.48	90.33 ± 4.14	79.65 ± 6.23	70.90 ± 4.22	86.76 ± 3.44	**91.33 ± 2.76**
6	48.67 ± 10.67	36.38 ± 6.86	27.52 ± 6.41	34.48 ± 5.81	10.95 ± 15.84	13.05 ± 7.07	**52.67 ± 10.33**	42.86 ± 9.80
7	**64.55 ± 8.41**	46.21 ± 8.26	46.98 ± 9.40	56.77 ± 5.95	45.96 ± 9.57	42.21 ± 5.77	54.64 ± 12.68	61.57 ± 9.75
8	68.61 ± 8.23	60.20 ± 11.55	61.75 ± 7.19	70.54 ± 5.84	62.62 ± 4.13	62.28 ± 6.11	67.80 ± 2.88	**71.03 ± 2.21**
9	78.91 ± 3.89	82.72 ± 3.47	82.12 ± 2.17	79.72 ± 2.14	82.51 ± 2.28	72.07 ± 4.82	82.91 ± 2.90	**83.13 ± 2.02**
10	81.98 ± 3.66	77.25 ± 2.63	78.68 ± 4.04	80.88 ± 4.24	81.24 ± 2.48	71.43 ± 5.95	**84.29 ± 3.37**	81.48 ± 3.02
11	60.42 ± 6.80	43.16 ± 13.53	56.89 ± 5.59	59.42 ± 7.69	59.74 ± 7.46	57.11 ± 5.50	**68.05 ± 6.17**	68.00 ± 8.44
12	92.58 ± 4.03	91.79 ± 2.35	90.13 ± 2.82	86.29 ± 5.35	90.66 ± 3.44	91.09 ± 2.89	92.36 ± 2.86	**92.75 ± 2.43**
13	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
14	**100.00 ± 0.00**	99.91 ± 0.12	**100.00 ± 0.00**	99.91 ± 0.12	99.91 ± 0.12	99.95 ± 0.09	99.95 ± 0.09	99.48 ± 0.67
OA (%)	80.63 ± 1.15	76.74 ± 0.80	78.78 ± 0.87	80.10 ± 0.39	78.18 ± 1.32	71.98 ± 0.98	82.25 ± 0.85	**82.85 ± 1.17**
AA (%)	74.26 ± 1.15	65.92 ± 2.28	66.82 ± 2.08	68.97 ± 1.77	64.21 ± 1.65	63.90 ± 0.47	74.17 ± 2.31	**75.04 ± 3.65**
$\kappa\times100$	76.96 ± 1.25	72.01 ± 1.08	74.54 ± 1.06	76.26 ± 0.44	73.81 ± 1.54	66.54 ± 1.06	78.76 ± 1.05	**79.51 ± 1.43**

the proposed method with a different number of transformer blocks to find the best settings for different datasets. The results of the experiment can be found in Table VI. Taking SA dataset for example, when the number of blocks is 2, 1, and 1 in stage1, stage2 and stage3, respectively, the proposed model achieves the highest OA value of 96.80% on the SA dataset. For CNNs, the number of layers of the network usually has a great influence on the effect of classification. However, we note that the proposed GAHT performs robustly for different settings from Table VI. We speculate that this is because the MHSA mechanism can compute the relationship between features well using fewer layers. We set the best block numbers reported in Table VI in the following experiments.

D. Comparison With SOTA Methods

The results of the proposed method are compared with the most widely used deep learning methods: a multiscale 3-D deep convolutional neural network (M3D-DCNN) [23], a CNN-based 3-D deep learning approach (3D-CNN) [25], a deep feature fusion network (DFFN) [19], a RSSAN [39], an attention-based bidirectional long short-term memory network (AB-LSTM) [40], a transformer-based backbone network named SpectralFormer (SF) [43], and a SSFTT [52].

The classification results of different methods can be found in Tables VII–X. As the results illustrate, our proposed GAHT obviously outperforms the other SOTA HSI classification methods based on deep learning over the four benchmark datasets. On the SA dataset, GAHT performs much better than others and achieves 2.12% higher than RSSAN and 3.07% higher than SSFTT in terms of OA. Also, the proposed GAHT outperforms other models in terms of AA and *kappa* values. Similar results can also be found at WHU-LK dataset, which GAHT leads other models in the three indicators of OA, AA, and *Kappa*. And on the PU dataset, GAHT shows its advantages in eight of nine categories. On the most challenging HR-L dataset, it is shown that two transformer-based methods (SSFTT and GAHT) perform much better than others. Based on this, we can conclude that vision transformers with modification have greater potential on datasets which are difficult to classify.

TABLE XI
OA (%) OF WITH ORIGINAL PATCH EMBEDDING AND GPE IN THE SA, PU, WHU-LK, AND HR-L DATASETS

Embedding Module	SA	PU	WHU-LK	HR-L
Original Patch Embedding	95.02	96.99	98.94	81.32
Grouped Pixel Embedding	**96.80**	**97.87**	**99.26**	**82.85**

TABLE XII
CLASSIFICATION RESULTS AND TOTAL TRAINABLE PARAMETERS OF PROPOSED METHOD WITH HIERARCHICAL AND NONHIERARCHICAL STRUCTURE ON THE SA, PU, WHU-LK, AND HR-L DATASETS

		SA	PU	WHU-LK	HR-L
OA (%)	Nonhierarchical	96.50	97.63	99.20	82.19
	Hierarchical	**96.80**	**97.87**	**99.26**	**82.85**
Params. (M)	Nonhierarchical	1.69	2.30	2.89	1.99
	Hierarchical	**0.97**	**0.93**	**1.51**	**0.82**

We choose the one with the highest OA values of five experiments to visualize the predicted samples of different methods to qualitatively compare these models. As shown in Figs. 6–9, the classification maps of GAHT have fewer noises than other methods on four benchmark datasets. It is observed that the proposed GAHT provides the result that is closest to the ground truth. To sum up, the GAHT generates the most accurate and smooth classification maps, which further indicates its superiority.

E. Ablation Study

1) Impact of the Proposed GPE Module: Table XI shows the effects of the proposed method with GPE and original patch embedding in transformers. Compared with ordinary patch embedding, the proposed GPE can better explore the unique characteristic of grouped features of feature maps.

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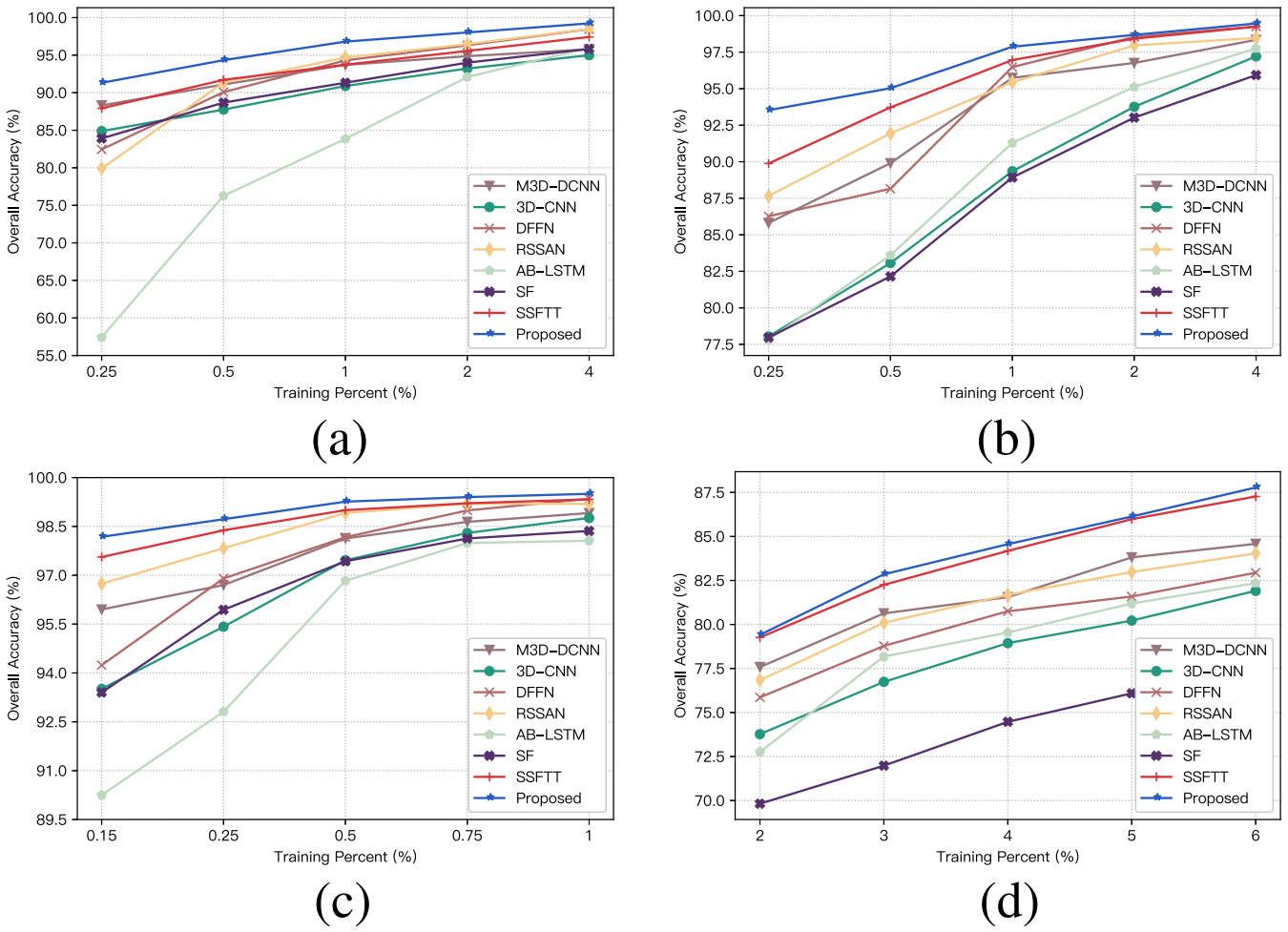


Fig. 10. Classification results versus different percentage of training samples on the four datasets. (a) SA. (b) PU. (c) WHU-LK. (d) HR-L.

Specifically, the OA of the transformer with the GPE is 1.78% higher than that with the original patch embedding in the SA dataset. A similar result is reported in the HR-L dataset, the increment of OA is 1.53%. As for the PU dataset, the GPE module leads to 0.88% growth in terms of OA. These fully reflect the superiority of the proposed GPE structure. However, since the original OA values are already relatively high, the gain brought by the GPE module on the WHU-LK dataset is limited to a certain extent.

2) Impact of Hierarchical Structure: To show the effects of the proposed hierarchical structure, the hierarchical and nonhierarchical transformers are compared, and the results over four datasets are shown in Table XII. The nonhierarchical transformer with a nonhierarchical structure is identical to our proposed model except it does not employ dimension reduction in grouped convolutions, which means the number of channels in them is fixed. It is observed that the transformer with a hierarchical structure achieves better accuracies with about only half parameters compared with its nonhierarchical counterpart.

F. Robustness Evaluation

To evaluate the robustness of the proposed transformer network, we test our method with other competitors along with the increasing number of training samples. Specifically, 0.25%, 0.5%, 1%, 2%, and 4% labeled samples were randomly chosen as training data for SA and PU datasets, while 2%, 3%, 4%, 5%, and 6% for the HR-L dataset, and 0.15%, 0.25%, 0.5%, 0.75%, and 1% for the WHU-LK dataset. Fig. 10 shows the results in terms of overall accuracies of different classifiers. We can observe the proposed method clearly outperforms the others under all cases and shows its robustness. Take the SA dataset for example, when the number of training samples is limited, the AB-LSTM generated inferior OA, while the GAHT still can achieve high classification accuracy. And for the HR-L dataset, it is found that the OA of our proposed algorithm can greatly outperform other methods except the SSFTT under different sampling ratios of training samples. For a large number of training samples in SA, PU, and WHU-LK datasets, because the classification accuracy is quite high, the improvement of the OA of the proposed model is not obvious compared to other models.

V. Conclusion

In this article, we investigate common problems in existing vision transformers for HSI classification. To tackle them, we propose our group-aware hierarchical transformer to classify HSIs. A novel embedding module call grouped pixel embedding is proposed to learn features within a relatively local area, which can handle the limitation of over-dispersion in MHSA. This results in generating spatial–spectral features in a global–local style and focusing on classification-related characteristics. Moreover, our transformer is a hierarchical

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network. Compared to the nonhierarchical transformer, transformer with hierarchical structure can greatly reduce parameters of the model. Meanwhile, the hierarchical structure gradually increases the semantic information in the feature map by gradually reducing the channel size of the feature map. Hence, the performance of classification is further improved. Experimental results over four benchmark HSI datasets have demonstrated the superiority of the proposed method, in terms of both visual qualities on the classification map and quantitative metrics.

In future work, we will further boost the ability of feature extraction of the network by refining the structure of MHSA mechanisms. Besides, we will try to embed the feature fusion module into the network to fuse our multiscale features generated by the hierarchical transformer.

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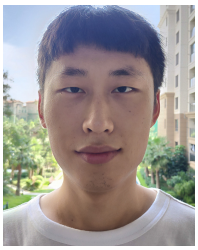
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